

JUUPAJOKI HYYTIALA, Finland

WMO: 027700

Lat: 61.846N

Lon: 24.287E

Elev: 502

StdP: 14.43

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.1	-7.4	-17.3	2.2	-12.8	-11.3	3.0	-7.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	80.3	63.9	76.5	62.5	72.9	60.8	67.3	75.4	65.1	72.5	62.9	70.1	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	92.9	69.6	62.3	85.5	68.0	60.1	79.1	65.6	32.0	75.9	30.3	72.1	28.7	70.3	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	N/A	N/A	N/A	DB	-16.6	83.5	6.9	4.8	-21.6	87.0	-25.7	89.8	-29.5	92.5	-34.6	96.0
(5)				WB	-17.0	68.8	6.8	3.8	-21.9	71.5	-25.9	73.7	-29.7	75.8	-34.7	78.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	39.7	17.6	21.3	27.0	37.5	49.3	56.0	61.7	58.5	50.2	39.0	32.4	24.9	(6)
(7)		DBStd	16.83	12.08	11.62	8.71	6.42	7.89	6.43	6.02	5.35	5.93	6.40	7.86	11.00	(7)
(8)		HDD50	4748	1005	804	714	378	111	15	1	3	69	343	527	778	(8)
(9)		HDD65	9287	1470	1224	1179	825	489	278	140	214	443	805	977	1243	(9)
(10)		CDD50	995	0	0	0	3	88	196	364	265	76	3	0	0	(10)
(11)		CDD65	60	0	0	0	0	1	9	39	11	0	0	0	0	(11)
(12)		CDH74	610	0	0	0	0	56	117	349	88	0	0	0	0	(12)
(13)		CDH80	108	0	0	0	0	3	17	74	14	0	0	0	0	(13)
(14)	Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
(15)	Precipitation	PrecAvg	23.6	1.8	1.3	1.2	1.2	1.5	2.7	3.1	2.8	2.1	2.3	1.9	1.8	(15)
(16)		PrecMax	33.1	3.5	2.7	2.5	2.4	3.5	4.2	5.6	6.4	5.6	5.2	4.7	3.8	(16)
(17)		PrecMin	16.5	0.5	0.1	0.2	0.1	0.2	0.8	0.9	0.2	0.4	0.1	0.2	0.5	(17)
(18)		PrecStd	3.4	0.8	0.7	0.6	0.6	0.8	0.9	1.3	1.5	1.4	1.4	1.0	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.9	41.6	49.4	66.0	79.9	82.4	86.0	81.9	69.3	55.7	49.8	44.4	(19)
MCWB			36.8	37.1	39.6	50.2	58.8	63.3	65.9	66.7	60.2	51.6	48.3	43.4	(20)	
2%		DB	35.6	37.7	43.5	59.2	74.6	77.4	82.0	76.0	65.0	52.8	46.5	39.9	(21)	
		MCWB	34.6	35.4	37.3	46.0	57.2	61.0	65.9	63.1	57.7	50.2	45.6	38.8	(22)	
5%		DB	33.5	35.6	40.1	53.6	70.1	73.0	77.8	72.1	61.8	51.0	44.0	37.4	(23)	
		MCWB	32.9	34.5	35.7	43.4	55.5	59.3	64.5	61.4	56.0	49.0	43.3	36.4	(24)	
10%		DB	32.3	33.8	37.4	49.4	65.2	69.2	74.0	68.7	59.2	48.7	41.6	35.7	(25)	
		MCWB	31.8	32.9	34.2	41.4	52.4	57.7	63.2	59.8	55.4	46.9	40.9	35.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.5	38.0	41.3	51.2	62.0	67.1	70.8	68.8	61.2	53.1	48.5	43.8	(27)
MDCB			38.0	40.2	46.1	61.9	73.4	75.5	78.9	77.7	66.5	55.0	49.3	44.1	(28)	
2%		WB	34.9	35.9	38.4	47.9	59.5	63.7	68.4	65.4	59.3	51.4	45.6	39.1	(29)	
		MDCB	35.3	37.2	42.1	57.1	70.1	72.9	76.6	72.1	63.1	52.3	46.6	39.9	(30)	
5%		WB	33.1	34.5	36.5	44.8	56.8	61.3	66.5	63.4	57.7	49.3	43.4	36.6	(31)	
		MDCB	33.5	35.5	39.2	52.1	66.5	69.7	74.3	68.9	60.6	50.4	43.9	37.1	(32)	
10%		WB	31.9	33.1	34.7	42.3	54.1	59.1	64.4	61.5	55.9	46.9	41.2	35.0	(33)	
		MDCB	32.3	33.8	36.9	48.2	64.0	67.0	71.8	66.7	58.3	48.5	41.6	35.5	(34)	
(35)	Mean Daily Temperature Range	MDBR	10.3	10.5	14.9	17.5	20.8	18.3	18.0	17.2	14.0	10.1	6.6	7.3	(35)	
MCDBR			6.1	9.1	16.4	26.6	29.4	25.6	23.8	22.9	18.1	12.6	8.0	7.2	(36)	
MCWBR			6.1	7.8	11.1	15.3	14.0	11.8	10.0	11.5	11.4	9.8	7.6	7.1	(37)	
5% WB		MCDBR	6.1	8.0	13.3	23.0	25.4	21.3	20.3	19.2	15.3	10.9	8.1	7.1	(38)	
		MCWBR	6.1	7.1	9.7	14.2	13.0	10.7	9.9	10.5	10.6	10.0	7.8	7.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.243	0.290	0.317	0.346	0.344	0.342	0.365	0.351	0.328	0.317	0.266	0.356	(40)	
taud		2.064	2.153	2.232	2.394	2.469	2.497	2.450	2.480	2.538	2.411	2.165	2.609	(41)		
Ebn at Noon		140	210	250	268	279	281	270	265	251	205	131	91	(42)		
Edh at Noon		12	22	29	30	30	30	31	28	22	17	10	7	(43)		
(44)	All-Sky Solar Radiation	RadAvg	67	245	712	1163	1567	1663	1592	1234	720	299	78	35	(44)	
(45)		RadStd	12	57	86	131	174	131	162	140	92	56	15	4	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (54 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	-238	N/A	N/A	N/A

Nomenclature: See separate page